Guidelines for UI Design

Navigation Design

The component that enables the user to navigate through the system

Also provides messages of success or failure of actions performed

Make it simple so that the user never really notices

Basic principles:

- Prevent the user from making mistakes
  - *i.e. do not display a command that cannot be used*
  - *i.e. confirm critical actions that will take place*

- Simplify recovery for the user when mistakes are made
  - *i.e. UNDO button*

- Use a consistent grammar order
  - *Object-Action order vs. Action-Object order*
    - (File ► Open vs. Open ► File)

Types of Navigation Controls

Language

- Command language—user types in a command to be executed i.e. SQL commands
- Natural language—system interprets the user's language i.e. MS Help (free form questions)

Menus

- User is presented a list of choices
- Comes in different forms (e.g., menu bars, popups, drop downs)

Direct manipulation (e.g., drag and drop)

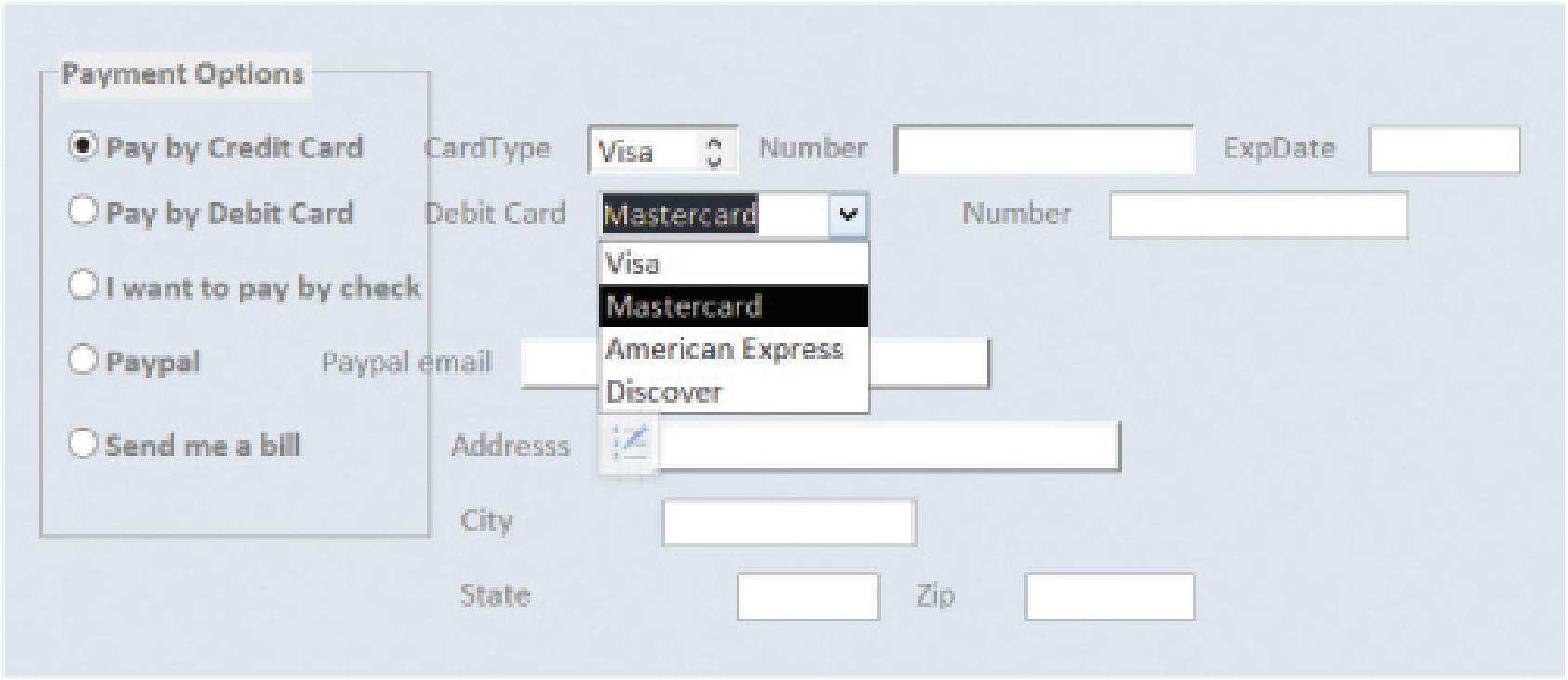
Messages

How the system informs the user of the status of an interaction

- Error messages—user did something that is not permitted
- Confirmation messages (e.g., “Are you sure?”)
- Acknowledgment messages (e.g., “Order entered”)
- Delay messages—provides feedback to the user that the process is running
- Help messages—provides additional information about the system to assist the user in performing a task

Layout and Formatting

- Design screens with a purpose
- Consider location and grouping
- Ensure professionalism



Data Entry

- Use control that minimise keystrokes - use dropdown lists, date pickers
- Set focus and use tabbing
- Set font and text box sizes appropriately
- Include online edits

Types of Input Controls

Free form controls

- Text boxes for alphanumeric information
- Number boxes with automatic formatting
  - *Example: Enter a phone number as 3451236789; automatically formats as (345)-123-6789*
- Password boxes that hide characters with stars (*) and do not allow cutting or copying

Selection boxes

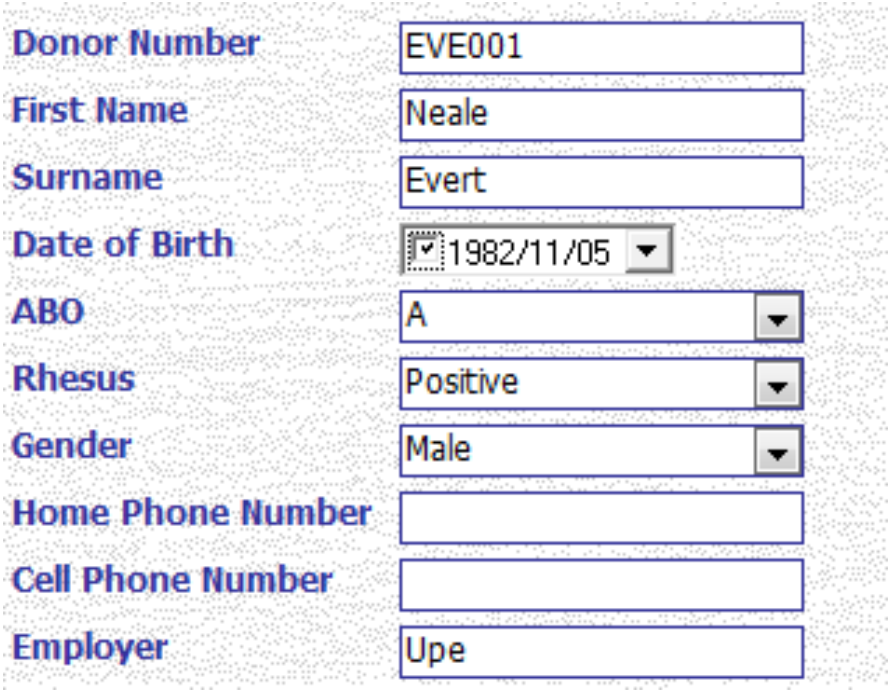
- Check boxes when several items can be selected
- Radio buttons when items are mutually exclusive
- List boxes to present a set of choices
- Sliders—a pointer that can be moved along a scale

Guidelines for Text Boxes

- Text boxes should be large enough to accommodate all the characters.

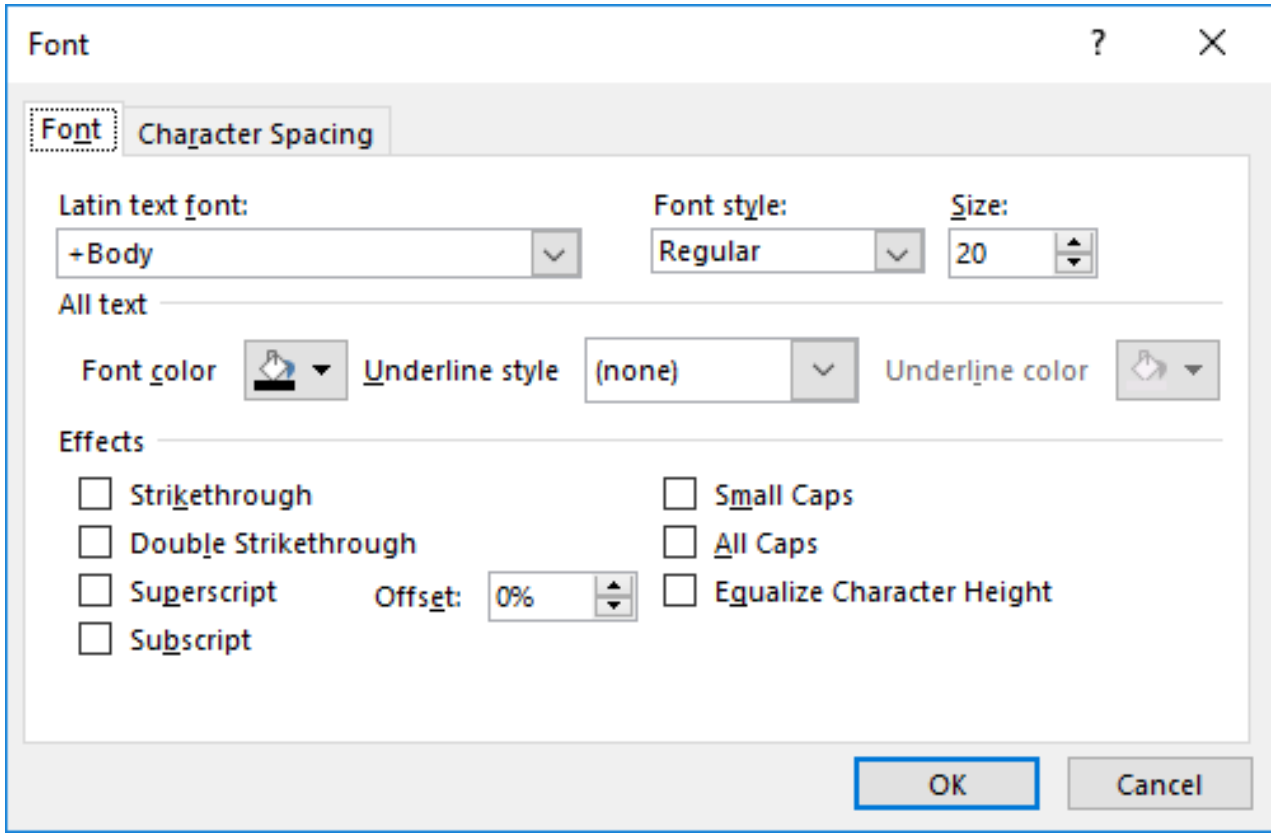


- Captions should be to the left of the text box.
- Character data is left-aligned within the box.
- Numeric data is right-aligned.



Guidelines for Check Boxes

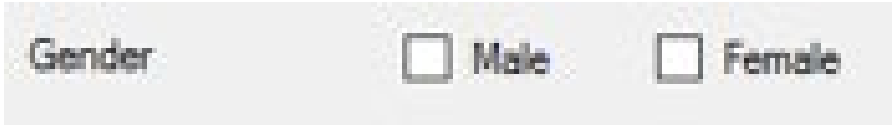
- Check boxes are used for non-exclusive choices.
- Check box text or label is placed to the right of the check box.
- If there are more than ten check boxes, group together in a bordered box.



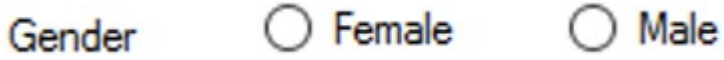
Guidelines for Option (Radio) Buttons

- Option or radio buttons are used for exclusive choices.
- Choices are listed to the right of the button, in some sequence.
- Often they are placed in a rectangle called an option group.
- If more than six option buttons are used, a list box or drop-down list box should be implemented.

X

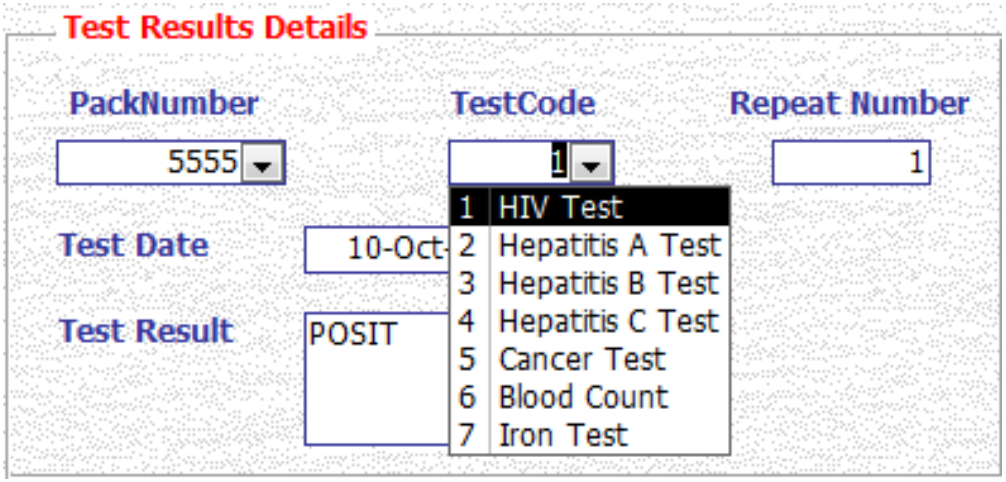


✓

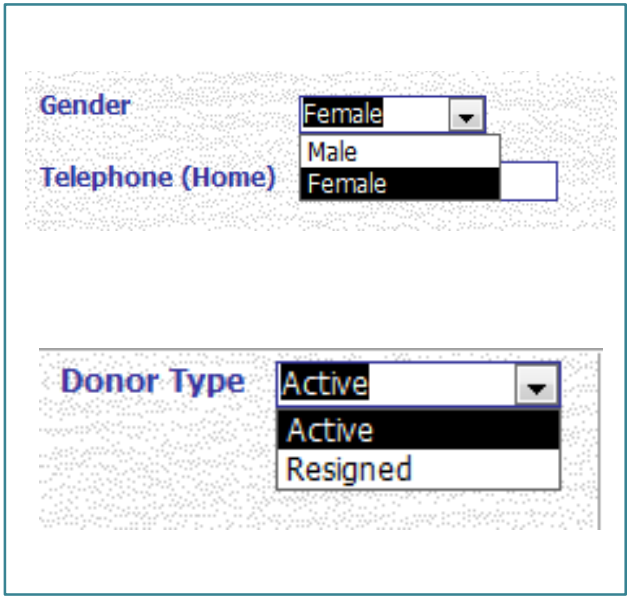


Guidelines for List and Drop-Down List Boxes

- Used when there is little room available on the page
- If there is a commonly selected choice, it is usually displayed in the drop-down list by default.

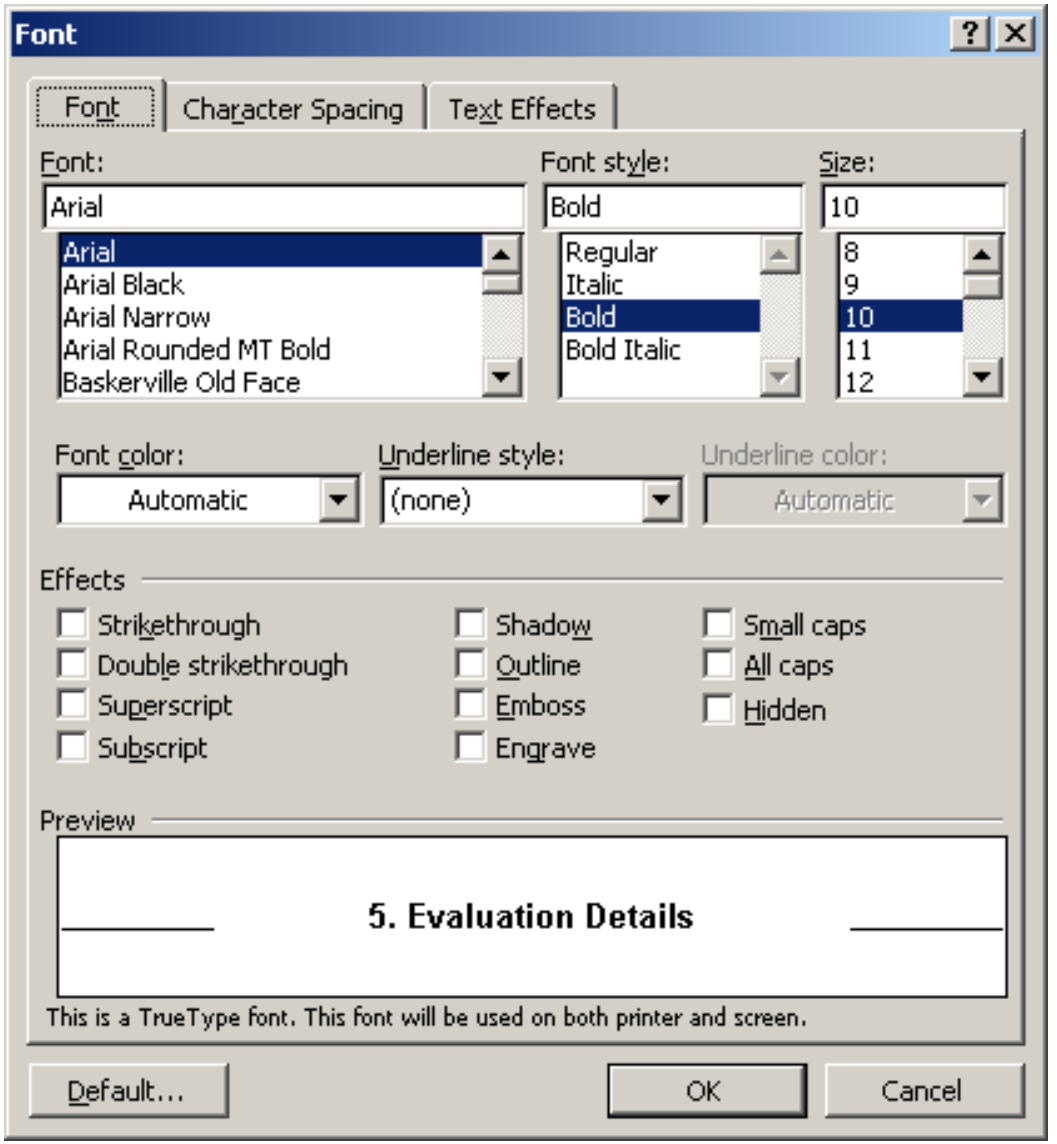


Incorrect use of list boxes



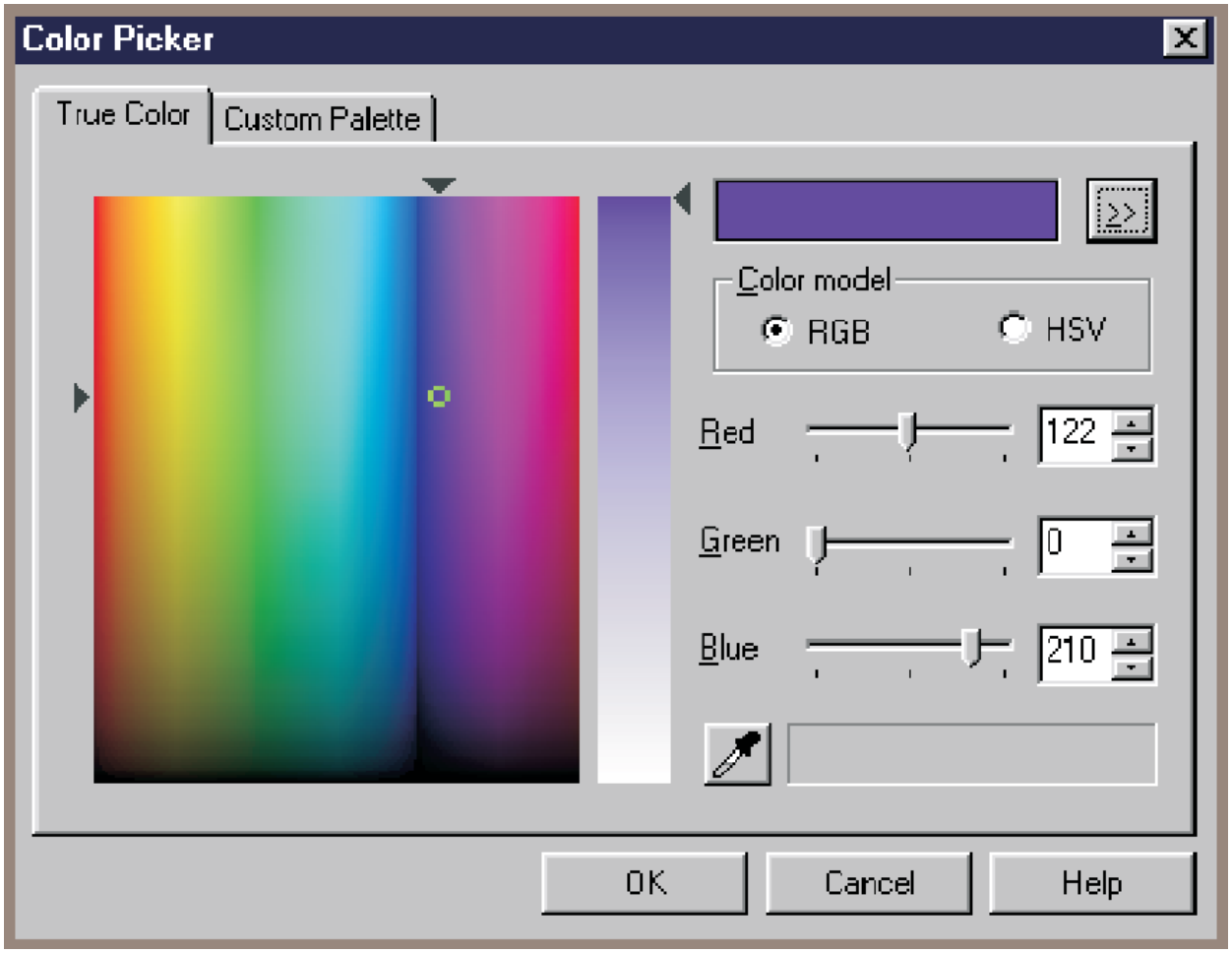
Guidelines for Tab Control Dialogue Boxes

- Create a separate tab for each unique feature.
- Place the most commonly used tabs in front and display them first.
- Place buttons outside of the tabs - user can correctly consider those buttons as controlling the entire set of tabs.



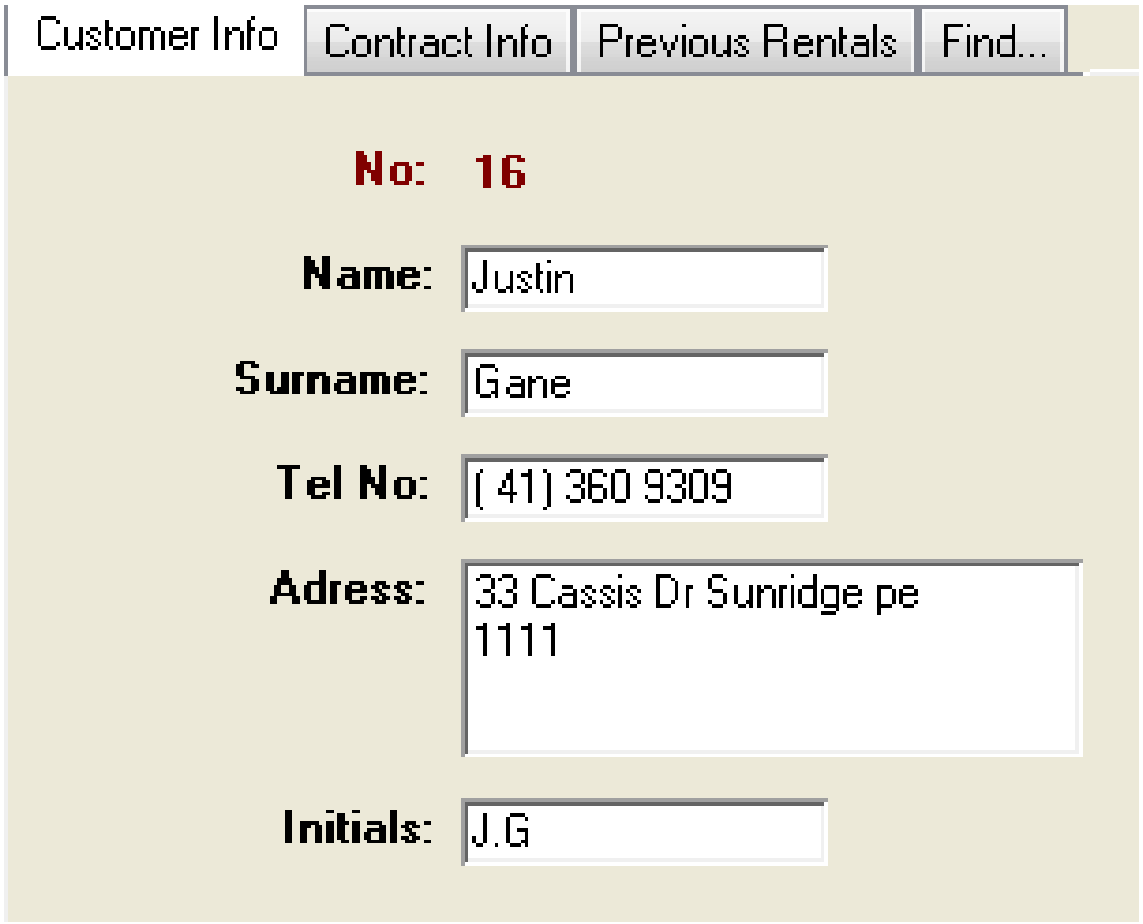
Guidelines for Sliders, Spin Buttons, and Image Maps

- Sliders and spin buttons are used to change data that have a continuous range of values.



Guidelines for Text Areas

- Text areas can include a number of rows, columns, and scroll bars that allow the user to enter and view text greater than the size of the box area.



Guidelines for Message Boxes and Command Buttons

- Message boxes are used to warn users and provide feedback messages in a dialog box.
- Command buttons perform an action when the user selects it.

